

Wine Quality Prediction

README “ GitHub Project Documentation”

This project is a part of the AAI-500 course in the Applied Artificial Intelligence Program at the University of San Diego (USD).

Project Status: In Progress **to be uploaded to GitHub later**

Installation

This project runs in Google Colab or any standard Jupyter environment. All data files are included in the repository.

Google Colab

The notebook can be opened directly in Google Colab by navigating to https://colab.research.google.com/drive/1wzfdjvHVafb5mPRJ4Xv8d57WPMnLF7n#scrollTo=m_cznUYi95. All cells run sequentially from top to bottom, with no additional configuration required.

Local Environment

The following command installs all required dependencies:

```
pip install pandas numpy matplotlib seaborn scipy statsmodels scikit-learn
```

The notebook is then launched with:

```
jupyter notebook TeamProject.ipynb
```

Project Intro / Objective

The main purpose of this project is to determine whether the physicochemical properties of Vinho Verde wine, obtained through standard laboratory testing, can reliably predict wine quality classifications assigned by human tasting panels. Expert tasting panels are expensive, time-consuming, and inherently subjective; a data-driven model offers a scalable and objective alternative for winery quality control.

This project applies a complete statistical pipeline connecting all seven modules of AAI-500 to a real-world business problem. The analysis demonstrates that two chemical measurements, alcohol content ($r = +0.44$) and volatile acidity ($r = -0.39$) drive the majority of the quality signal. The Bayesian Network shows that combining high alcohol with low volatile acidity raises the posterior probability of good quality classification to 88.6%. The GLM achieves AUC-ROC of 0.8045 on the held-out test set, and the Random Forest classifier achieves the highest accuracy at 78.3%.

Partners / Contributors

Name	Role	Sections
Tena Alshaikhli	Team Lead	Introduction, Statistical Analysis, Bayesian Network, GLM
Roopleen Kaur	Data and Model Analysis	Data Cleaning, EDA, Model Analysis
Timothy Hays	Modeling and Testing	Hypothesis Testing, Model Selection, Conclusion

GitHub: <https://github.com/Denaalshaikhli/Wine-Quality-Prediction>

Instructor: Professor Leon Spanner, University of San Diego

Methods Used

- Descriptive Statistics and Data Collection (Module 1)
- Sampling Distributions and Central Limit Theorem (Module 3)
- Inferential Statistics and Hypothesis Testing (Module 4)
- Bayesian Network Probabilistic Modeling (Module 5)
- Generalized Linear Models with Logit Link Function (Module 6)
- Machine Learning Classification: Random Forest, Gradient Boosting, Logistic Regression, Naive Bayes (Module 7)
- Machine Learning Regression: Linear Regression, Bayesian Ridge (Module 7)
- Data Visualization
- Data Manipulation and Preprocessing

Technologies

- Python 3
- pandas
- numpy
- matplotlib
- seaborn
- scipy
- statsmodels
- scikit-learn
- Google Colab
- Jupyter Notebook
- GitHub

Project Description

Dataset

Property	Value
Source	UCI Machine Learning Repository
Citation	Cortez et al. (2009). Decision Support Systems, 47(4), 547-553
Original samples	6,497
After cleaning	5,320
Red wine samples	1,359
White wine samples	3,961
Features	11 physicochemical variables
Target variable	Quality score 0-10, binarized at threshold 6
Good quality (score ≥ 6)	3,332 samples (62.6%)
Poor quality (score < 6)	1,988 samples (37.4%)

Research Question

Can physicochemical properties of wine, measurable through standard laboratory testing, reliably predict whether a wine will be rated as good or poor quality by human tasters?

Key Results

Analysis	Key Finding
Descriptive Statistics	Alcohol: Good 10.929% vs Poor 9.913% Volatile Acidity: Good 0.309 vs Poor 0.403
Central Limit Theorem	SE = 0.1677 from 5,000 samples of n=50 Confirms normality of sampling distribution
Confidence Intervals	Alcohol Good [10.888, 10.969] vs Poor [9.875, 9.951] — non-overlapping
Bayesian Network	High Alcohol + Low VA = 88.6% Low Alcohol + High VA = 17.3%
GLM (Logistic Regression)	AUC-ROC = 0.8045 Accuracy = 0.7415 Recall = 0.8367
Best Classifier	Random Forest: Accuracy = 0.783 F1 Score = 0.780

Data Dictionary

Variable	Description	Unit	Type
fixed acidity	Non-volatile acids	g/dm3	Continuous
volatile acidity	Acetic acid — vinegar smell at high levels	g/dm3	Continuous
citric acid	Freshness and flavor	g/dm3	Continuous
residual sugar	Sugar remaining after fermentation	g/dm3	Continuous
chlorides	Salt content	g/dm3	Continuous
free sulfur dioxide	Prevents microbial growth	mg/dm3	Continuous
total sulfur dioxide	Total SO2 (free and bound)	mg/dm3	Continuous
density	Mass per volume	g/cm3	Continuous
pH	Acidity level on 0 to 14 scale	—	Continuous
sulphates	Antimicrobial and antioxidant additive	g/dm3	Continuous
alcohol	Alcohol percentage by volume	%	Continuous
quality	Sensory score 0 to 10 — TARGET	—	Discrete

Challenges

- Coordinating a shared Colab notebook across three team members while maintaining section consistency
- The GLM was restructured to train on training data only and evaluate on the held-out test set to ensure proper generalization assessment
- Multicollinearity between density, alcohol, and residual sugar caused an inflated negative coefficient for density in the GLM
- The dataset is limited to Vinho Verde wines from Portugal and may not generalize to other wine varieties or regions

License

This project was created for academic purposes as part of the AAI-500 course at the University of San Diego. The UCI Wine Quality Dataset is publicly available under the Creative Commons Attribution 4.0 International license. Full license details are available at <https://archive.ics.uci.edu/dataset/186/wine+quality>

Acknowledgments

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References

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